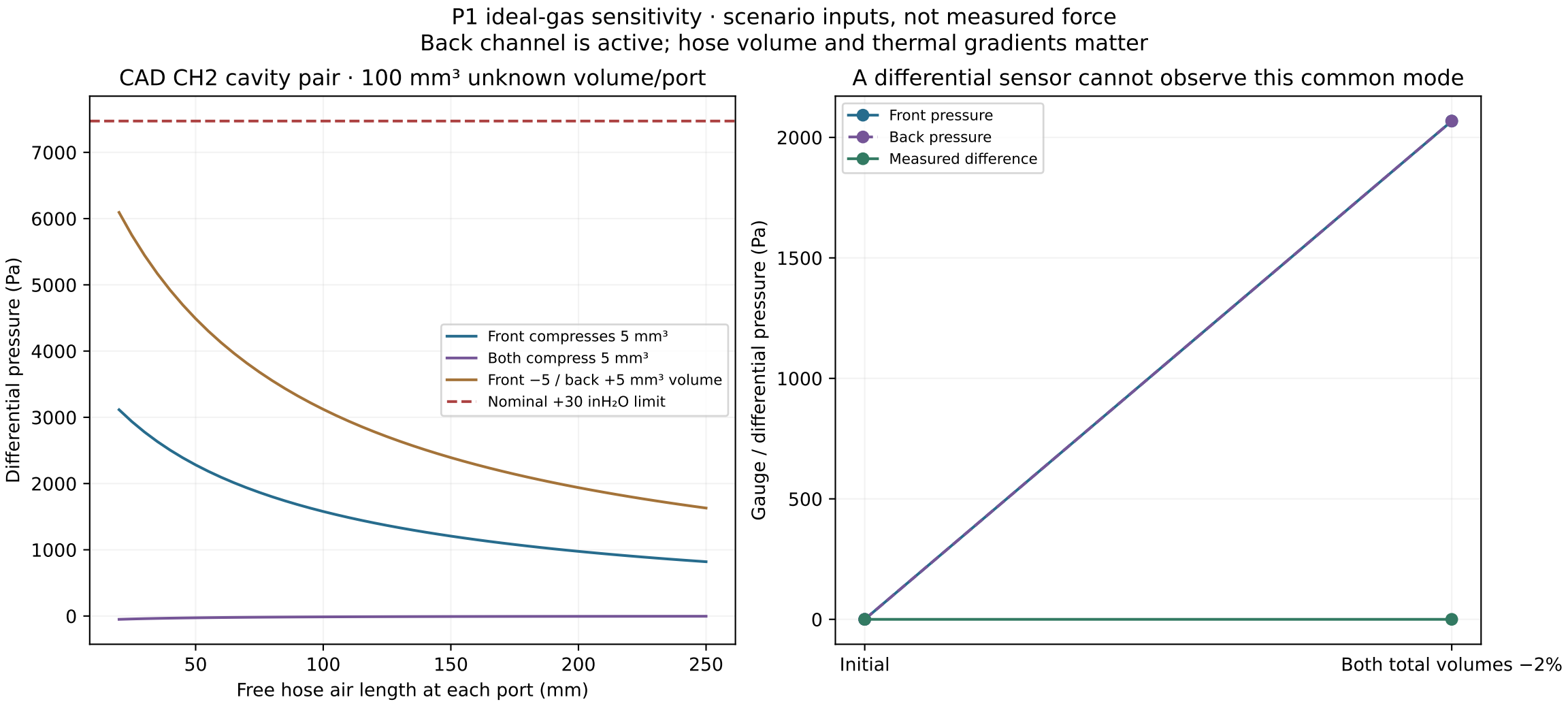
P1 ideal-gas sensitivity · scenario inputs, not measured force
Back channel is active; hose volume and thermal gradients matter
CAD CH2 cavity pair · 100 mm³ unknown volume/port
Differential pressure (Pa)
Free hose air length at each port (mm)
Front compresses 5 mm³
Both compress 5 mm³
Front −5 / back +5 mm³ volume
Nominal +30 inH₂O limit
A differential sensor cannot observe this common mode
Gauge / differential pressure (Pa)
Front pressure
Back pressure
Measured difference
Initial
Both total volumes −2%